

THE COSMETIC-SURGERY MANUSCRIPT IN CLASSICAL LANGUAGE

Labels: [V] means checked by hand or by script; [P] means plausible; [S] means speculative. $Y_r = S_r^3(K)$, $g(K) = 2$, and $\phi: Y_{-2} \rightarrow Y_2$ is orientation preserving. DLME means Daemi–Lidman–Miller Eismeier, arXiv:2410.21248.

0. Summary

The manuscript is DLME’s ± 1 argument transplanted to ± 2 , with one substitution: *Chern–Simons energy positivity is replaced by a spin-coupled positivity*.

- DLME. A filtered isomorphism $I(Y_1) \rightarrow I(Y_{-1})$ lowers $\ell = \min(\text{CS} - \text{gr}/8)$ by $\frac{1}{8} + \eta$. Combined with ϕ , this is a contradiction.
- The obstacle at ± 2 . Every available map $Y_2 \leftrightarrow Y_{-2}$ raises that level (§2), as DLME’s closing remark predicts.
- The fix. Couple the level to a spin^c structure. Each instanton A carries the coupled Dirac index $n_D(A) = \Theta - E(A) + \rho(\alpha) - \rho(\alpha')$. The $\text{PU}(2)$ -monopole cobordism (Pidstrigach–Tyurin localization) shows that, when no Seiberg–Witten-type reducibles are present, instanton counts with $n_D \geq 1$ vanish.
- The consequence. Maps are then filtered by $\sigma_t = \text{CS} - \rho_t$, and $b^+ = 1$ pieces lower it.

The stack plus repetition is the closed-manifold form of “iterate a level-lowering automorphism until the level gap is exceeded”. The manuscript works in closed form so that it never needs a $\text{PU}(2)$ Floer theory on $Y_{\pm 2}$.

1. The ordinary layer is DLME/CDX surgery-triangle machinery

Theory: Floer’s irreducible complex with determinant-one gauge. $C(S^3) = 0$; $Y_{\pm 1}, Y_{\pm 2}$ have only central reducibles; Y_0 carries w with $w \cdot A$ odd.

Manuscript	Classical counterpart	New
$f_+ : Y_0 \rightarrow Y_1 \rightarrow Y_2$; mirror f_-	Floer’s triangle twice; the S^3 term is 0 (DLME Prop. 3.6)	QHS ends with central reducibles
$g_+ : Y_2 \rightarrow S^3 \rightarrow Y_0$, interval to the $\mathbb{RP}^3$ split, two lifts	DLME’s g_1 (Prop. 3.7, §5.5) for the distance-two pair $\{2, 0\}$ (that is, ± 1 for the framing $\lambda + \mu$). The middle term is $S^3 \oplus S^3 = 0$; the $\mathbb{RP}^3$ ends cancel between lifts.	Y_0 admissible, not a ZHS
pentagons, $f_+ g_+ \simeq \text{id}$	DLME/CDX h -maps and triangle detection	—
$B : Y_2 \rightarrow Y_{-2}$, interval to the $L(4, 1)$ split, sphere test $x(S)$	distance-four analogue of g_1 . Mod 2, $B \simeq g_- g_+ = f_+^{-1} f_-^{-1} = H^{-1}$ [V].	<i>integral</i> lift (lens signs)
$H = f_+ f_-$	composite of Floer-triangle maps through Y_0 , with $b^+(W_H) = 1$	—

Why B must be integral, not just mod 2 [V]. The conclusion is $2^\eta \Omega = 0$, which is vacuous mod 2 once $\eta \geq 1$. The cap pairing q is only known to be a nonzero integer, possibly even. Hence the integral B and the mod- 2^N repetition: every step is a unit of $L/2^N L$, where $L = I(Y_2; \mathbb{Z})/\text{Tor}$. Repetitions divisible by $\exp \text{GL}_r(\mathbb{Z}/2^N)$ give $\Omega \equiv \pm q \neq 0$.

2. Why DLME’s filtration stalls at ± 2

Notation: for a count of index $i_0 = \text{codim} - \dim G$, the degree is $D = -2c^2 - 3b^+ + \dim G - \text{codim}$ and the level is $L = -c^2/4 - \eta$. DLME’s lemma then gives $\ell(Y') \leq \ell(Y) + (L - D/8)$ for injective maps.

map	data	$L - D/8$ (CS)	$L_\sigma - D/8$ (spin, §3)
DLME g_1 (± 1)	$c^2 = -1$, $\dim G = 1$	$-\frac{1}{8} - \eta$	$-\frac{1}{8}$
B (either lift)	$c^2 \in \{0, -4\}$, $\dim G = 1$, $\text{codim } 2$, $\Theta = 0$	$+\frac{1}{8} - \eta$	$+\frac{1}{8}$
H	$b^+ = 1$, $\Theta = 1$	$+\frac{3}{8} - \eta$	$-\frac{5}{8}$
period: k copies of B and one H		$\frac{k+3}{8} - \eta_{\text{tot}}$	$\frac{k-5}{8}$

These values are [V] (formal index bookkeeping; script). With Chern–Simons alone, no combination is favourable without a quantitative bound on η . This agrees with DLME’s closing remark on ± 2 .

3. The spin-coupled level, and where $\text{PU}(2)$ enters

Fix a spin^c structure with determinant l , put $\Lambda = l + c$, and let $\Theta = (\Lambda^2 - \sigma)/4$. APS gives, for A on W from α to α' , with $E(A) = -c^2/4 + \text{CS}(\alpha) - \text{CS}(\alpha')$,

$$n_D(A) = \Theta_W - E(A) + \rho_t(\alpha) - \rho_{t'}(\alpha'),$$

where $\rho_t(\alpha)$ is the $(\eta + h)/2$ -correction of the 3-dimensional Dirac operator coupled to $\alpha \otimes t$ [P].

- *Spin monotonicity.* If every counted A has $n_D(A) \leq 0$, then $\sigma_t := \text{CS} - \rho_t$ drops by at least $c^2/4 + \Theta_W$, so the spin level is $L_\sigma = -c^2/4 - \Theta_W$. This is DLME’s mechanism with η replaced by Θ .
- *The period.* By the table, the spin shift per period is $(k - 5)/8$. This is exactly minus the manuscript’s $\Delta n_D = (5 - k)/8$, coming from $n_D = \frac{5m-n}{8} + \text{const}$ [V]. So the lower window bound $m/n > 1/5$ is exactly “the period strictly lowers the spin level”.
- *What supplies the monotonicity.* $\text{PU}(2)$ localization: the instanton links are $\mathbb{CP}^{n_D-1}$, and the $\eta = n_D - 1$ phase cuts are $\mathcal{O}(2)$ -sections, giving 2^η . These links are cobordant to the reducible (SW-type) fixed strata plus face terms. With no reducibles in the window, the counts with $n_D \geq 1$ vanish.
- *This is a property of blocks, not of single maps* [V]. Mod 2, $HB \simeq \text{id}$, yet its spin shift is $-\frac{1}{2}$. Per-map monotonicity would give $\ell_t(Y_2) < \ell_t(Y_2)$ for every nontrivial K . So the reducible exclusion must fail on short blocks. It holds only when each positive cell is surrounded by enough (-4) -spheres (§4), and *this is the only place ϕ is indispensable*: without ϕ , every B must be followed by H .

Floer-theoretic form (new theory, resembling predecessors) [S]. Define spin-filtered IP-modules $I_\bullet^t(Y)$, filtered by $\sigma_t - \text{gr}/8$, with invariant ℓ_t . This requires ϕ^* -compatibility of t on $Y_{\pm 2}$, where $H^2 = \mathbb{Z}/2$. The monotonicity input would be an S^1 -equivariant $\text{PU}(2)$ -monopole Floer theory whose u -adic filtration is the n_D -level, with localization onto instanton plus SW fixed points. Compare Seidel–Smith/Hendricks–Lipshitz–Sarkar localization and Frøyshov- or Daemi- Γ -type invariants. The manuscript’s closed stack is the device that avoids this theory: all Y -seams stay finite, and only the psc necks S^3 and $L(4, 1)$ become infinite.

4. The reducible exclusion as basic-class constraints

Type S is an SW solution for $K = \Lambda + v$, perturbed by $\beta^+ \in 2\pi\Lambda$. Type R is an Abelian ASD reducible, simultaneously an SW wall and a Donaldson/Kotschick–Morgan wall. The level condition $\kappa_i = -v_i^2/4 + \ell_i$ with $\ell_i \geq 0$ is Feehan–Leness’s [V].

Manuscript (family, local)	Classical counterpart
positive cell $P \cong S^2 \times S^2 \# 4\overline{\mathbb{CP}}^2 \cong \mathbb{CP}^2 \# 5\overline{\mathbb{CP}}^2$ for every K [V]; $l_P^2 = 4 = 2\chi + 3\sigma$	homological SW profile of a rational surface
Weitzenböck: $(vH)(KH) < 0$; vertex classes $H_I = U + \frac{1}{4} \sum_{i \in I} X_i$	$b^+ = 1$ wall-crossing with $s \geq 0$; H_I is the period after rationally blowing down the X_i ; the clamp is convexity over the period polygon [V]
long $S^2 \times S^1$ tube forces $v \cdot U = 0$	neck-stretching proof of adjunction for a square-zero sphere [V]
reference fillings with $ K \cdot S_i \leq 4$; lens Dirac table	adjunction for (-4) -spheres and its reflection pair $K \pm 2\text{PD}(S)$ (Fintushel–Stern) [P]
test costs: $v \cdot S_i \neq 0$ or one unit of ℓ	$\mu(S_i)$ on a reducible link is proportional to $\langle v, S_i \rangle$ [P]
signed sum over 2^n bits, lens ends cancel	Fintushel–Stern rational-blowdown projection onto $K \cdot S_i \equiv 2 \pmod{4}$ [P]
caps: no R; $v_W^2, K_W^2 \leq C_W$; large odd $\Lambda(E_{\text{exc}})$	generic periods; finiteness of basic classes; blow-up formula

Window [V], by recount. For a block of k intervals with one bridge:

- $\Delta n_D = (5 - k)/8$;
- the fixed-type (S/R) score is $\geq 3k - 7$, so $k \geq 3$ suffices for it;
- the score of mixed P/R runs is $\geq k - 4$, so $k \geq 4$ is needed.

So spacing four is needed only for the mixed lemma (`indices:mixed-run`). Counting the forgotten R-wall condition might allow $k = 3$ [S].

5. Closed manifolds give nothing, and that is the point

- The stack is a connected sum along the $J_i \cong S^3$: $X \cong Z_- \# N_\phi^{\#(n-1-m)} \# P^{\#m} \# Z_+$, with $N_\phi = X_{-2}(K) \cup_\phi (-X_2(K)) \approx 2\overline{\mathbb{CP}}^2$. Hence $D_X \equiv 0$ [V].
- Ω is a *secondary* family invariant. Its fixed-metric virtual dimension is $-n$, and the cube’s faces are the connected-sum and lens degenerations [V].
- Rationally blowing down all S_i removes the connected-sum structure. The bit signs become the Fintushel–Stern projection. But B is not the blown-down cobordism map, and $D_{X_B} = 0$ by Witten’s conjecture [P].

So no closed-manifold Donaldson theorem replaces §§6–10. The needed input is the family/relative $\text{PU}(2)$ localization of §3 together with the exclusion of §4.

6. What is new, minimally stated

- (1) The integral negative interval B , with the lens-wall sign comparison that carries over unchanged to the coupled problem.
- (2) *Family localization* (“Theorem R”). Take a cube of metrics whose faces stretch psc rational homology spheres, with $n_D \geq 1$, and suppose the only fixed points are instantons and every mixed tuple has negative projected index. Then $2^{n_D-1}\Omega$ equals the sum of the face terms. Everything except the “necessary projection” for mixed P/R tuples (`analysis:projection`) adapts Feehan–Leness I/II, Morgan–Mrowka–Ruberman and Kotschick–Morgan [P].
- (3) The lattice budget (§8 of the paper), i.e. the window $\frac{1}{5} < m/n \leq \frac{1}{4}$, which is ϕ ’s only essential use.

Sanity test [P]. The argument seems to use ϕ only as a $b^+ = 0$ cobordism $Y_{-2} \rightarrow Y_2$ that is a mod-2 unit. If so, it also forbids any negative-definite $V: Y_{-2} \rightarrow Y_2$ inducing an $\mathbb{F}_2$ -isomorphism on I (for nontrivial K). That is a DLME-style “no ribbon/definite cobordism” statement and a natural standalone theorem. A single counterexample would refute the method.